CMPS 350 – Web Development Fundamentals

Practical Midterm Exam

Instructions:

- The exam duration is **120 minutes**. Read the questions carefully and plan your time accordingly.
- Push your code to **GitHub** regularly (at least every 30 minutes) to avoid unpleasant surprises.
- The exam is **open book**. In case of plagiarism, both parties will receive **0 points**. **Do not share or receive any code.**
- **Using AI tools (such as ChatGPT, Copilot, or any AI-powered code generator) is strictly forbidden.**
- Once you complete the exam, you should:
  - Add a screenshot for each question to the provided testing sheet.
  - Push your **code and testing sheet** to your GitHub repo under the `midterm` subfolder.
  - Demo your work before leaving the exam.

Cryptocurrency Wishlist Tracker

In this exam, you will apply your **frontend web development skills** to build a **cryptocurrency wishlist tracker**. You will fetch data from an API, allow users to store cryptocurrencies in a wishlist, and implement sorting and display functionalities. The complete working web application will be demonstrated at the start of the exam.

Implementation Tasks:

1. Fetch and Display Cryptocurrency Data [30 pts]

- Retrieve cryptocurrency prices from the following API: CRYPTO API

- Display the following details for each coin in a **grid layout** as shown in figure 1.
  - **Name**
  - **Symbol**
  - **Price (USD)**
  - **Market Cap**
  - **Icon**
- Ensure that the grid layout is **fully responsive**.

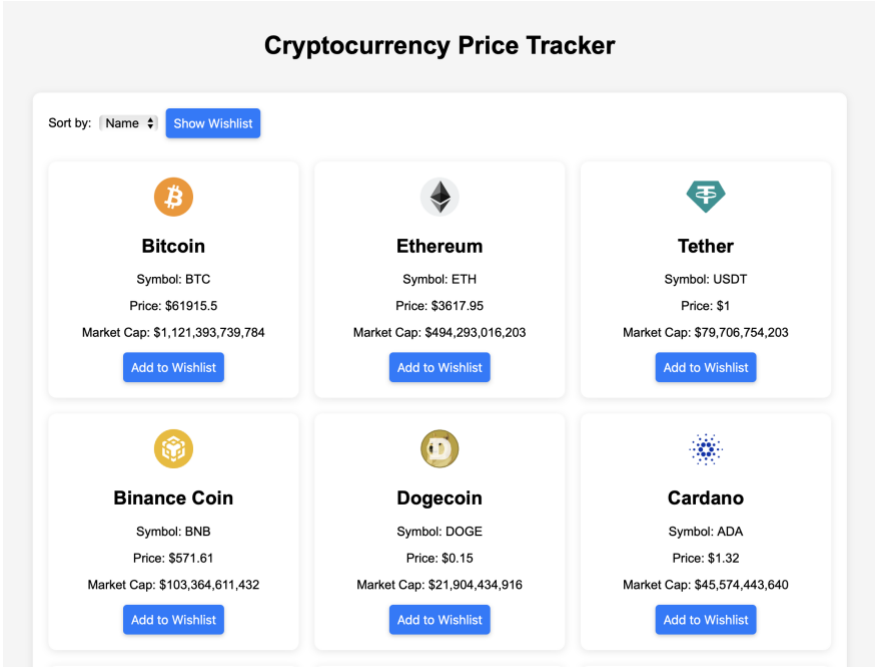

Figure 1 Cryptocurrency Wishlist Tracker

2. Implement Sorting [20 pts]

- Provide a **dropdown menu** to allow sorting by:
  - **Name** (Alphabetical order)
  - **Price** (Low to high)
- Ensure that when the user selects a sorting option, the cryptocurrency list is re-rendered accordingly.

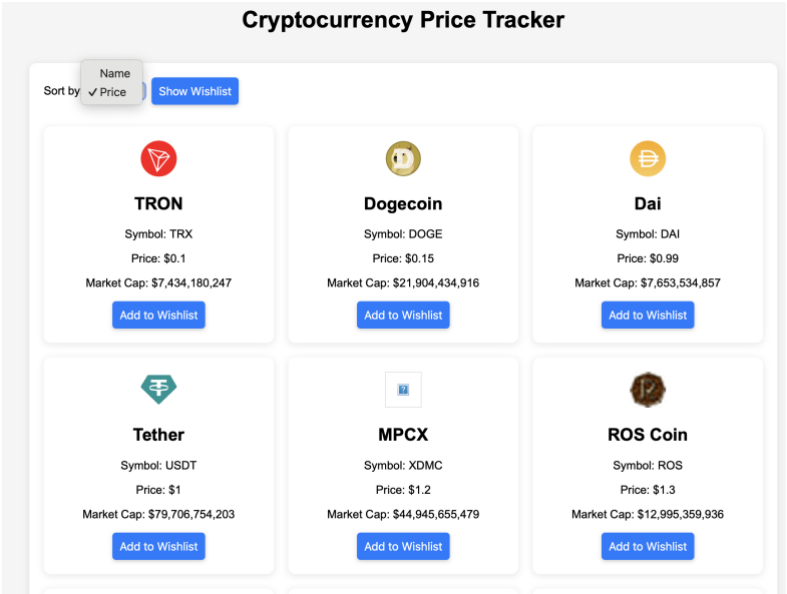

Figure 2 : After sorting by price

3. Implement a Wishlist Feature [25 pts]

- Add an **"Add to Wishlist" button** for each cryptocurrency.
- Clicking the button should store the selected coin's data in `localStorage`.
- If a user tries to add a coin that is already in the wishlist, an **alert should be displayed** informing them that the coin is already in the wishlist.
- Provide a **"Show Wishlist" button**, which loads and displays all saved wishlist coins from `localStorage`.

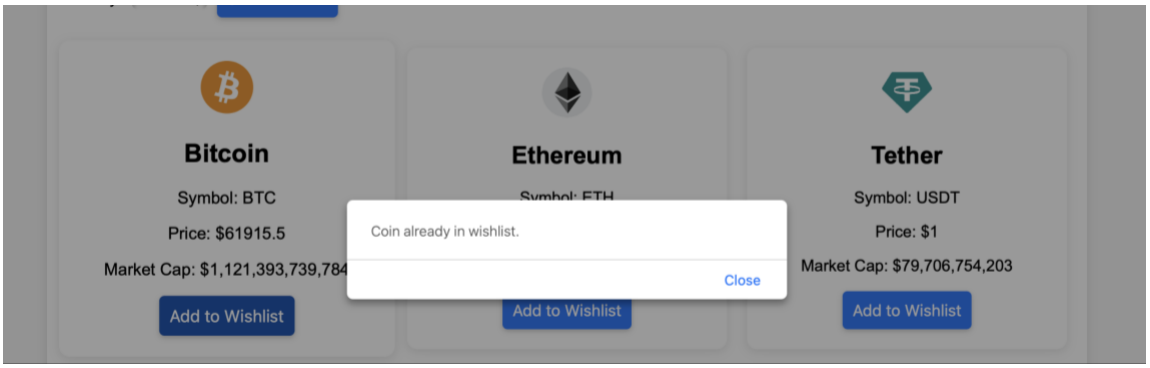

Figure 3 : Adding to wishlist an already exiting coin

4. Implement Wishlist Management [25 pts]

- When a user views the **wishlist**, they should be able to:
  - **See all previously saved cryptocurrencies.**
  - **Remove a cryptocurrency from the wishlist.**
- Ensure that removing a cryptocurrency updates the localStorage and the displayed wishlist accordingly.

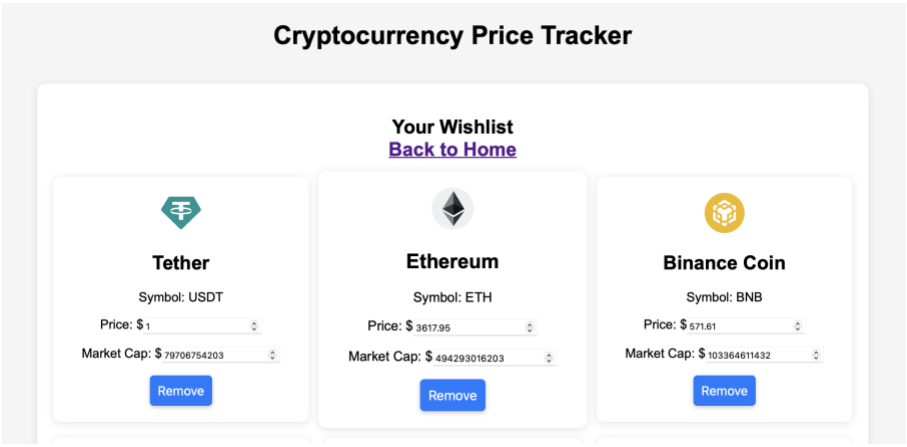

Figure 4 : Implement Wishlist Management

Grading Breakdown:

Task	Points
Fetch and display cryptocurrency data	30 pts
Implement sorting functionality	20 pts
Implement a add to wishlist feature	20 pts
Implement wishlist management (viewing & removing items)	30 pts
Total	**100 pts**

Good luck!